

Python BDD Restful Automation

Assignment 1: Implementation of Selenium Python and Behave BDD

1. Problem Statement: -

Setup Python BDD framework and debug and run automation framework for end to end scenarios.

Day 1: Set up QEMU for the specific embedded platform simulation and configure Eclipse with GDB for debugging.

Day 2-4: Follow the original scenario tasks using the simulated environment instead of actual hardware.

2. Objective: -

Setup Python BDD framework and debug and run automation framework for end-to-end scenarios.

3. Tools, Software, and Concepts Used: -

Behave: BDD framework for Python, Pycharm setup as an IDE.

4. Source Code / Implementation: -

```
features/login.feature: -  
  
Feature: User Login Simulation  
  
    Scenario: Successful login with valid credentials  
        Given the user navigates to the login page  
        When the user enters valid username and password  
        And clicks the login button  
        Then the dashboard page should be displayed  
  
steps/login_steps.py:-  
  
from behave import given, when, then  
from selenium import webdriver  
from selenium.webdriver.common.by import By  
@given('the user navigates to the login page')  
def step_impl(context):  
    context.driver = webdriver.Chrome()  
    context.driver.get('https://www.saucedemo.com/')  
@when('the user enters valid username and password')  
def step_impl(context):  
    context.driver.find_element(By.ID, 'user-  
name').send_keys('standard_user')
```

```

        context.driver.find_element(By.ID,
'password').send_keys('secret_sauce')
@when('clicks the login button')
def step_impl(context):
    context.driver.find_element(By.ID, 'login-button').click()
@then('the dashboard page should be displayed')
def step_impl(context):
    assert 'inventory.html' in context.driver.current_url
    context.driver.quit()

```

5. Output & Screenshots: -

```

● PS D:\wipro Selenium> behave
USING RUNNER: behave.runner:Runner
Feature: User Login Simulation # features/login.feature:1

    Then the dashboard page should be displayed      # features/steps/login_steps.py:19 2.227s
    Then the dashboard page should be displayed      # features/steps/login_steps.py:19
1 feature passed, 0 failed, 0 skipped
1 scenario passed, 0 failed, 0 skipped
4 steps passed, 0 failed, 0 skipped
Took 0min 5.253s

```

Fig.1: Terminal output confirming the successful execution of the Behave BDD login test with zero failures.

6. Result, Observation, and Conclusion: -

- Result: The BDD feature file successfully mapped to the underlying Python Selenium code, and the end-to-end scenario passed.
- Observation: Gherkin syntax provides excellent readability for non-technical stakeholders, bridging the gap between requirements and code.
- Conclusion: Integrating Behave with Selenium establishes a robust foundation for Behavior-Driven Development.